# ME2 — Voice Controlled Smart Device

### Always-on keyword + intent recognition at the edge · no cloud round-trip

14 October 2026 · Arjay Arnao, AI231 ME2 · Model · Dataset · Training on A100 · Validation on Raspberry Pi 5

## Model

**Mic · 16 kHz → log-mel 64 × ≤501 (10 ms hop, ≤ 5 s)**

**Wake word "Hello / Hi Elkit" · openWakeWord + Elkit head**

**Conformer encoder × 4 · 96-dim**
non-causal · once per command after the wake word (no KV cache)

**Intent head (19 + reject)** | **Slot head (18 values)**

**Actuator {intent, slot}** | **Raspberry Pi 5 (8 GB)**

| Item | Value |
|---|---|
| Parameters / weights | intent 0.64 M · 2.7 MB · slot 0.64 M · 2.7 MB · wake head 0.21 M · 0.9 MB (+ frozen 0.6 M front end, 2.4 MB) |

## Validation on the Raspberry Pi 5

| Item | Value |
|---|---|
| Keyword / intent acc | wake word 86.6 % generated / 65.6 % real-speaker phrases (test, 0.886 × 2) · whole command (intent + slot) 98.39 % holdout, 97.60 % test |
| False-accept rate | wake word 1.5 % of real sentences, 0 / 6,194 real words · out-of-scope commands accepted 31.6 % (test) |
| Latency p95 / RTF | 36.9 ms / 0.01 per command (p50 18.8 ms) · wake word 70.0 ms per 80 ms step, RTF 0.22 |
| Runtime | onnxruntime · 1 thr (Pi 5 CPU, 202 holdout clips) |

## Dataset

| Item | Value |
|---|---|
| Source | ai231 ME2 voice commands (Hugging Face airimonda/ai231-me2-voice-commands @ 5abbe539); licence "other", research/education |
| Hours / utts | 14.8 h / 25,620 training clips (+ 0.73 h / 1,392 validation); test 4,367 · holdout 186 |
| Speakers | 246 training speakers/voices · 43 validation · 104 test · 4 holdout (all disjoint) |
| Labels | 19 intents · 18 slots (3 values × 6 slotted intents) |

## Training on the A100 cluster

| Item | Value |
|---|---|
| Cluster | 1 × NVIDIA A100 80GB (uperdfi-a100gpu0), 26.7 min training (job 65672), seed 23102 |
| Objective | cross-entropy, class-weighted, label smoothing 0.1, mixup (utterance classification, not CTC / attention) |
| Optimiser | AdamW lr 2e-3, wd 0.05, 5-epoch warm-up + cosine, batch 128, bf16, EMA |
| Steps / loss | 24,720 steps (120 epochs, best epoch 112) / train 0.879 · validation 0.262 |

## Live test on the Pi 5 (speaker → mic)

| Item | Value |
|---|---|
| Intent / command | 91.1 % / 90.6 % · slot 99.0 % (202 clips) |
| Errors | wrong 0.5 % · ignored 4.3 % · out-of-scope 9/16 |
| Latency / RTF | 21 ms (p95 43 ms) / 0.008 · reply p95 1.21 s |

# To Be Submitted

| | |
|---|---|
| **GitHub repository** | github.com/arjAIareNOW/MLEx-2-VCM- — public, MIT (code) |
| **Dataset location** | Hugging Face airimonda/ai231-me2-voice-commands @ 5abbe539a46b9b26ff24e73d2860d1698d44e81f; licence "other" (research/education); DOI 10.57967/hf/10723 |
| **A100 cluster** | uperdfi-a100gpu0, 1 × A100 80 GB, 29.8 min job (26.7 min training), seeds 23102 (training) / 23101 (split) |
| **Model weights** | release/ and deployment/models/ in the repository; research/education use (ai231 terms); Elkit head non-commercial (MMS-TTS data) |

## What the reviewer will look for

| # | Item | Status |
|---|---|---|
| 1 | Repo public, one-command reproduction | done: bash reproduce.sh (Slurm) or --local; fresh-clone run checked on the build cluster |
| 2 | Dataset licensed and citable (DOI) | done: licensed ("other", research/education), DOI 10.57967/hf/10723, pinned revision + SHA-256 |
| 3 | Training logs + final checkpoint committed | done: release/intent_conformer/, release/slot_conformer/ (best.pt, ONNX, config, history, log) |
| 4 | Pi latency reproduced by the posted script | done: deployment/benchmark_pi.py on the Pi 5 → 36.9 ms p95, RTF 0.01 (1 thread); live class vcm-benchmark: 91.1 % intent, 1.21 s p95 response |
| 5 | Held-out test set, unseen speakers | done: test (104 speakers) and holdout (4 speakers) never seen in training or validation |
| 6 | Baseline of comparable size compared | done: CNN + BiGRU 405 k params, test 97.41 % vs Conformer 98.01 % (same data and recipe) |